

EECE 5554 Robotics: Sensing and Navigation**Syllabus: Section 01 (CRN: ECE 41116; ME 35561)**

This course examines the mathematical techniques and practical methods for sensing and navigation in mobile robots with a focus on sensors such as inertial measurement units (IMUs), Global Positioning Systems (GPS) and cameras. Course topics include basic properties of sensors, inertial sensors, GPS systems, error analysis, the Kalman filter, sensor fusion, Dead Reckoning, SLAM (simultaneous localization and mapping) and systems aspects of robotics. A large component of the course involves programming in ROS (Robot Operating System) to obtain real-world data sets using GPS, RTK GPS and IMUs for determining the path and estimating the pose of a mobile system. The course culminates with a team-based final project focused on the analysis of robot sensor data and/or the implementation of a navigation algorithm.

Prerequisite(s): MATH 3081 with a minimum grade of D- or EECE 3468 with a minimum grade of D-; and EECE 2160 with a minimum grade of D- or EECE 2210 with a minimum grade of D- ; or permission from the graduate program and the instructor.

Meeting Place and Times:

Snell Library, Room 035

Mondays, Wednesdays, and Thursdays, 09:15 AM – 10:20 AM

Instructor: Dr. Thomas R. Consi

Teaching Professor, Dept. of Electrical and Computer Engineering

Office: Dana 405

Email: t.consi@northeastern.edu

Office Hours: Thursdays, 12:30pm – 2:30pm, or by appointment.

Course Assistants:

Rohith Poola, poola.r@northeastern.edu

Office Hours: Tuesdays, 5:30pm – 6:30pm, Room: Ryder 273

Ruchik Jani, jani.ru@northeastern.edu

Office Hours: Fridays, 6:00pm – 7:00pm, Room: Dana 440

Srijan Dokania, dokania.s@northeastern.edu

Office Hours: Wednesdays, 6:00pm – 7:00pm, EXP 7th Floor Kitchen Area or EXP 730

Online Tools

Piazza: We will use Piazza as our primary means of communication and dissemination of information (lecture notes, lab handouts, assignments, etc.).

Sign-up Link: <https://piazza.com/northeastern/spring2025/eece5554>

Link to Piazza Page: <https://piazza.com/northeastern/spring2025/eece5554/home>

GitHub: GitHub will be used for handing-in assignments and for code development. We will instruct you on creating and using a GitHub account in the first lecture.

Canvas: The EECE5554 Canvas page will be used primarily for grading and for submitting reports. <https://canvas.northeastern.edu/>

ROS wiki: The main source of information about ROS, the Robot Operating System. Includes a good beginner's tutorial to get you up and running with ROS.

We will use ROS 2 Humble Hawksbill distribution

<https://docs.ros.org/en/humble/>

Student Groups: In the first week of the semester, you will form a group of four students that will stay together for the labs, the journal club and the project. Students are permitted to change groups as long as they inform Dr. Consi.

For a week-by-week schedule of topics and labs see the *"EECE 5554 Schedule, Section 01, Spring 2025"* in the Resources section of Piazza.

Course Deliverables

Labs: You will work individually for Lab 0. For Labs 1 – 5 you will gather your data together (unless otherwise specified) and use the same data for your reports. However, each student within the group must perform their own analysis and write their own lab report. These will be graded individually.

Journal Club: It is important for you to learn how to critically read research papers in robotics in the relevant journals. Each group will select a journal article of their choice and present it to the class. It is recommended that you pick a journal article that is relevant to the topic that you will explore for your project.

Project: The project will be a substantial piece of work involving sensing and navigation. Project deliverables: proposal, presentation, written report. The project is a group activity and all members of the group will receive the same grade.

Grade Breakdown:

Labs 0, 1, 2, 3, and 5: 10% each, 50% total

Labs 4: 15%

Journal Club: 10%, group grade

Final Project: 25% total, group grade

Project Proposal: 5%

Presentation: 5%

Report: 15%

Late Policy: you must submit your work the date it is due. Late assignments will be penalized according to the following Python formula:

$$\text{finalScore} = \text{onTimeScore} * (1 - \text{hrsLate} // 24 / 10)$$

Grading Scheme: Every deliverable will be graded out of 100. The percentages will be adjusted in Canvas. Numerical grades with fractions will be rounded up to the nearest integer grade.

A	≥ to 93%
A-	90% to 92%
B+	87% to 89%
B	83% to 86%
B-	80% to 82%
C+	77% to 79%
C	73% to 76%
C-	70% to 72%
D+	67% to 69%
D	63% to 66%
D-	60% to 62%
F	≤ 59%

Course Policies

Missing Class:

We will be taking attendance at each class. This is to allow us to monitor your participation and let us know if you are having problems with the class.

If you know in advance that you will miss a class meeting, you are expected to submit any relevant assignment on time through GitHub and Canvas.

COVID Policy:

Please read the latest Northeastern COVID-19 policy,
<http://news.northeastern.edu/coronavirus/>

Masking is optional for attending classes, it is up to you to determine your own comfort level.

Contact Dr. Consi ASAP if you come down COVID or have any other health problem that will cause you to miss class and/or deadlines for deliverables. We will make sure that you can continue your course work remotely and/or make-up for lost time.

Academic Integrity:

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

Northeastern University (and the instructional staff of this course) expect students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors. <https://osccr.sites.northeastern.edu/academic-integrity-policy/>

Please read this about cheating: <http://www1.coe.neu.edu/~jaaisaacs/honesty.html>

Academic Accommodations:

Northeastern University's Disability Access Services (DAS) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university.

We strive to make the class as accessible as possible. If you have documentation on file with the DAS and you believe those accommodations apply to this course, then email Dr. Consi to set up a meeting to discuss course policies and assignments and make sure that your accommodations are being met.

If you feel you need academic accommodations but do not have appropriate documentation on file with the DAC, then contact them at: <https://disabilityaccessservices.sites.northeastern.edu/>

Title IX:

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance. Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff. In case of an emergency, please call 911. Please visit <https://www.northeastern.edu/ouec/title-ix-policy-2/> for a complete list of reporting options and resources both on-and off-campus

We will seek to keep information shared with me as private as possible, but we are obligated to report any situations relating to sexual misconduct or harassment that we become aware of to the Title IX office, which may result in the office contacting you to provide you with resources and ask for further information.